UniEvents

Mobile Computing Project

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1. Introduction

University campuses are vibrant hubs of activity, yet the communication and discovery of events are often fragmented across multiple channels like emails, physical posters, and social media. This decentralization leads to low student awareness and engagement. UniEvents is a mobile application designed to solve this problem by providing a single, interactive platform for discovering, managing, and navigating campus events. By leveraging a map-based interface, the application aims to enhance the campus experience for both students and event organizers, fostering a more connected and engaged university community.

2. Problem Statement

The core problem UniEvents addresses is the inefficient and scattered nature of event communication within universities. This creates several pain points:

- **For Students:** Difficulty in discovering events relevant to their interests, keeping track of schedules, and finding event locations on a large campus.
- **For Organizers:** Challenges in reaching a wide student audience, managing event promotions, communicating updates, and handling attendee logistics effectively.

The result is missed opportunities for student participation and a diminished sense of community, as valuable events go unnoticed.

3. System Architecture and Workflow

The UniEvents application is built on a modular architecture using React Native and Firebase. The workflow is logically divided into four main modules as illustrated in the system diagram.

Module 1: Authentication and User Setup

This module handles user identity and access control.

- Users, whether they are students or organizers, sign in to the application using their Google accounts.
- Firebase Authentication is used to securely manage the login process, integrating with the Google API to verify credentials.
- Upon successful authentication, a unique UserID is generated and used throughout the application to manage user-specific data, such as event enrollments and created events.

Module 2: Student Registration (Event Discovery)

This is the primary interface for students to find and engage with events.

- Students can browse a list of current and upcoming events.
- The application features an interactive React Native map that displays event locations as markers, providing a visual and intuitive discovery experience.
- Students can select an event on the map to view details and apply or enroll in it directly through the app.

Module 3: Organizer Board (Event Management)

This module provides a dedicated dashboard for event organizers to create and manage their events.

- Organizers can add new events or update existing ones.
- Key details such as the venue, date, time, and description can be managed.
- A camera module is integrated, allowing organizers to upload relevant images like event posters, flyers, or pictures of the venue to provide more context to students.

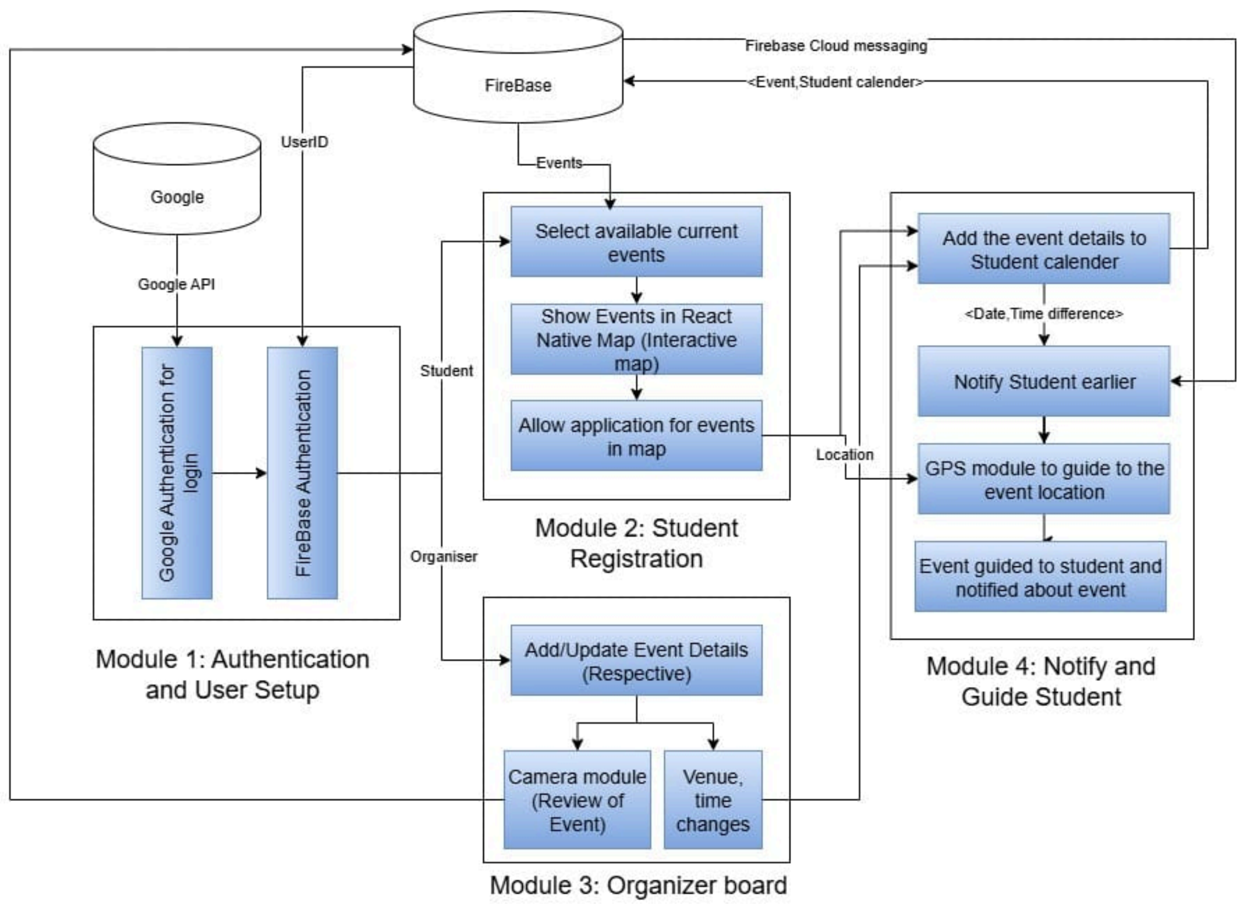
Module 4: Notification and Guidance

This module focuses on post-enrollment communication and logistics.

- When a student enrolls in an event, it is automatically added to their personal student calendar.
- Firebase Cloud Messaging (FCM) is used to send timely notifications and reminders to enrolled

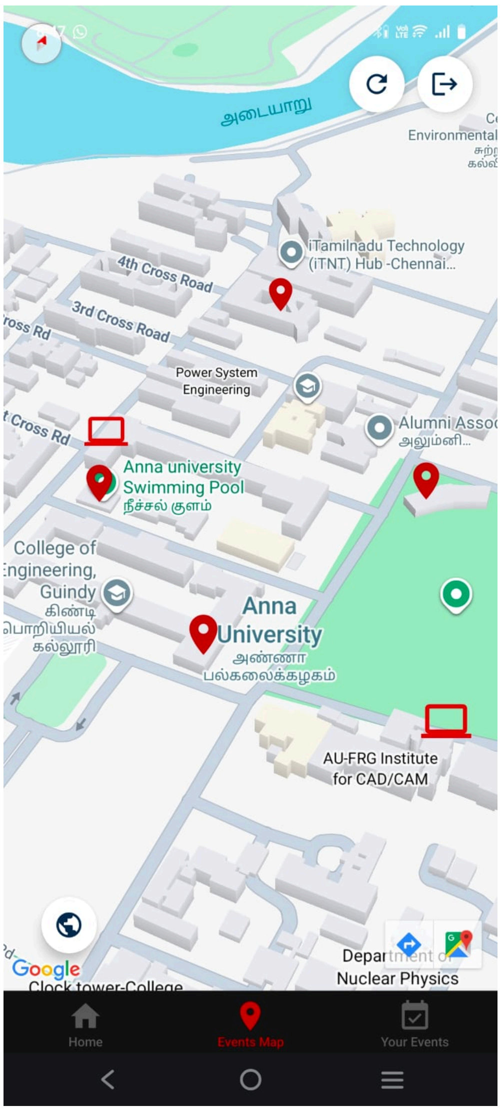
students, ensuring they don't miss the event.

- For on-campus navigation, the application integrates a GPS module to provide students with turn-by-turn directions to the event location.

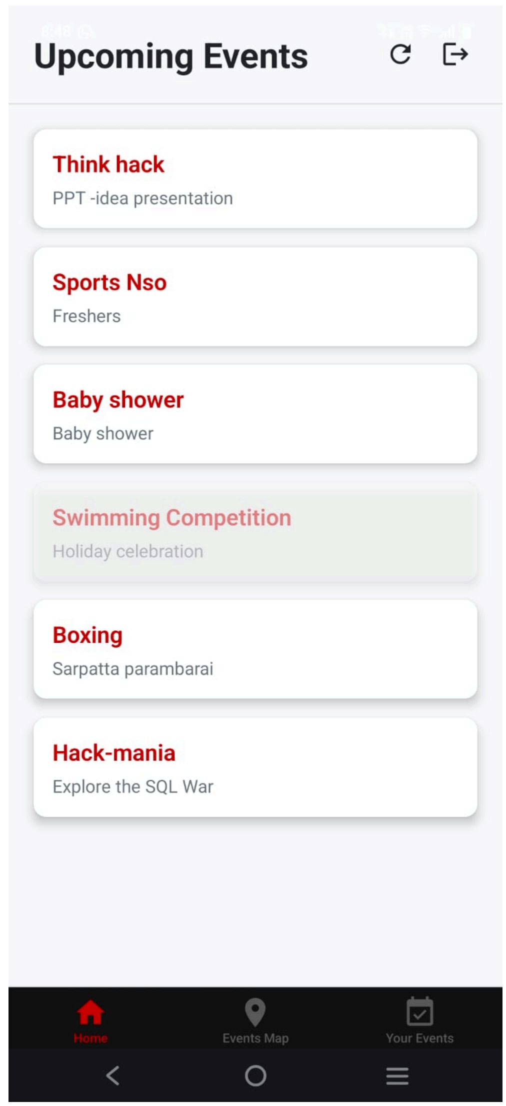


4. Key Features

- **Interactive Campus Map:** A real-time map interface showing all event locations.
- **Centralized Event Discovery:** Students can easily view details for all ongoing and upcoming events in one place.



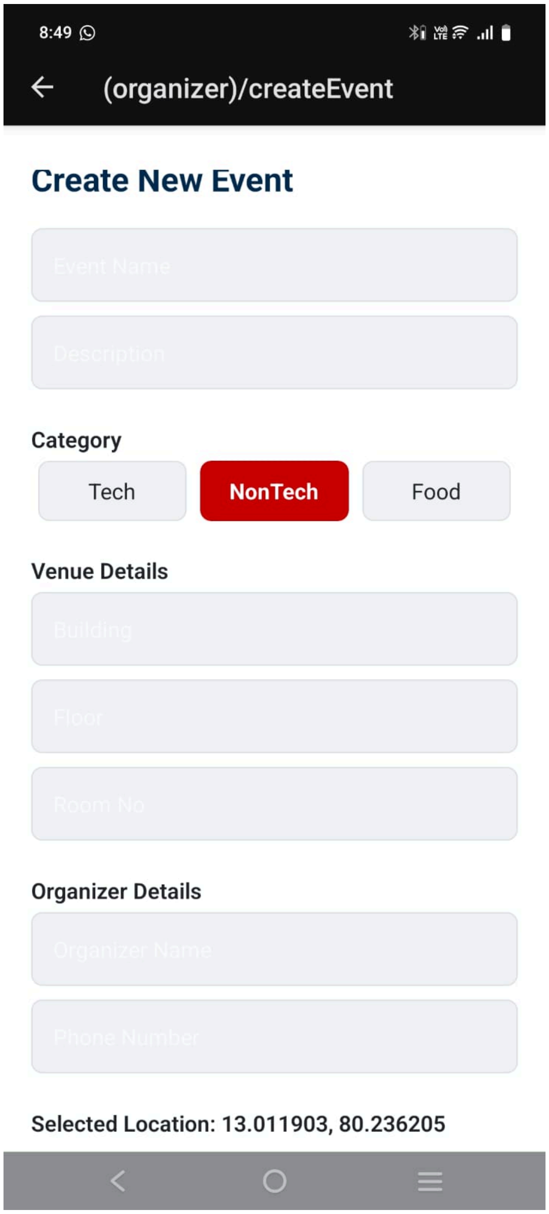
InterActive Map Feature

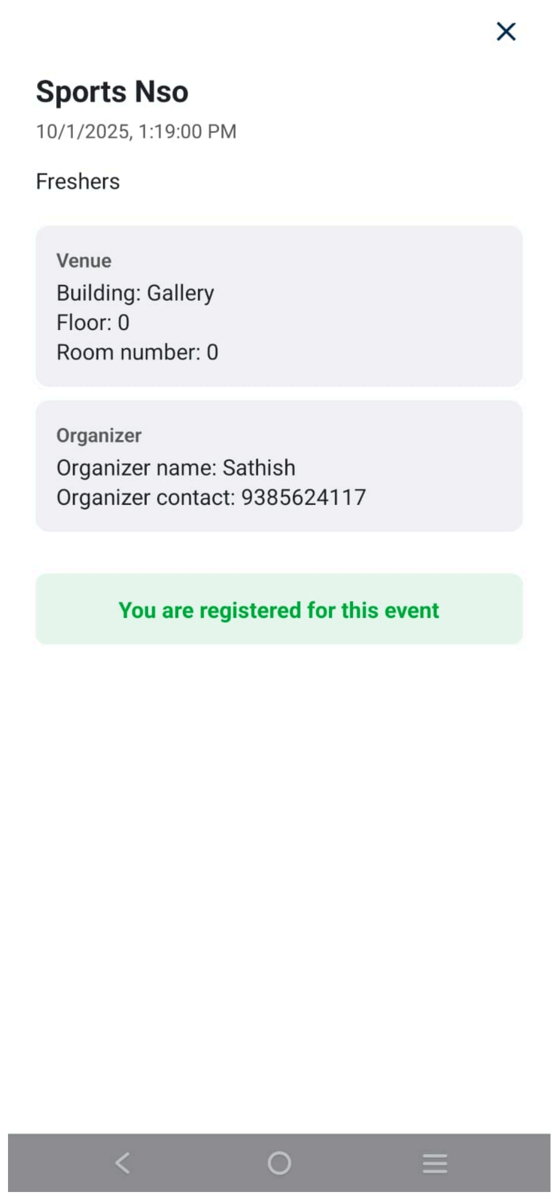


Event Discovery

- **Seamless Enrollment:** A simple one-tap process for students to enroll in events, which are then added to their personal calendar.
- **Automated Notifications:** Timely reminders for upcoming events and real-time updates from

organizers are sent via push notifications.





Event Creation **After Event Registration**

- **GPS-Powered Navigation:** In-app guidance to help students find their way to event venues.
- **Intuitive Event Creation:** A simple interface for organizers to add events by placing markers directly on the campus map.
- **Rich Media Attachments:** Organizers can upload photos (posters, venue images) to make their event listings more appealing.

- **Comprehensive Event Management:** A dashboard for organizers to manage all event details, including title, description, time, and attendee capacity.

5. Technology Stack

The technology stack for UniEvents is chosen for its efficiency, scalability, and suitability for creating a modern, cross-platform mobile application.

- **Frontend (Mobile App):** React Native is used to build the application for both iOS and Android from a single codebase. Its component-based architecture is ideal for creating a dynamic and interactive user interface, especially the map feature.
- **Backend and Database:** Firebase, a Backend-as-a-Service (BaaS) platform from Google, provides all the necessary backend functionality:
    - Firebase Authentication: For secure user login and management via Google Sign-In.
    - Firestore/Realtime Database: To store and sync all application data in real-time, including user profiles, event details, and enrollment lists.
    - Firebase Cloud Messaging (FCM): To power the push notification system for event reminders and updates.
    - Firebase Storage: To store user-generated content, such as the images and media uploaded by event organizers.

6. Conclusion

UniEvents presents a robust and user-friendly solution to the persistent problem of fragmented event communication on university campuses. By providing a centralized, map-based platform, it bridges the gap between event organizers and the student body. The application not only simplifies event discovery and management but also enhances the overall campus experience by fostering greater participation and community engagement. Its modern tech stack ensures a scalable, reliable, and maintainable system poised to become an indispensable tool for any university.